Outline of Absolute System

When you compose an absolute system using an absolute encoder, you are not required to carry out homing operation at the power-on, and this function suits very well to such an application as a robot.

Connect the host controller with the Minas A5 with absolute specifications. (motor with absolute encoder and driver with absolute spec) and set up the parameter, Pr0.15 to 0 or 2, then connect the battery for absolute encoder to compose an absolute system with which you can capture the exact present position information after the power-ON.

Shift the system to origin once after installing the battery and clear the multi-turn data by clearing the absolute encoder, then you can detect the absolute position without carrying out homing operation.
Via RS232 or RS485 communication, the host controller can connect up to 32 MINAS-A5 and capture the present position information as serial data to obtain the absolute position of each axis by processing. each data.

Applicable Mode

You can use all of MINAS A5 series driver in absolute specifications by setting up parameter. Use the motor which 8th place (designated for rotary encoder specifications) is “S” (7-wire type).

M ∗ M ∗ ∗ ∗ ∗ S ∗ ∗ ∗ ∗

8th place — Rotary encoder specifications

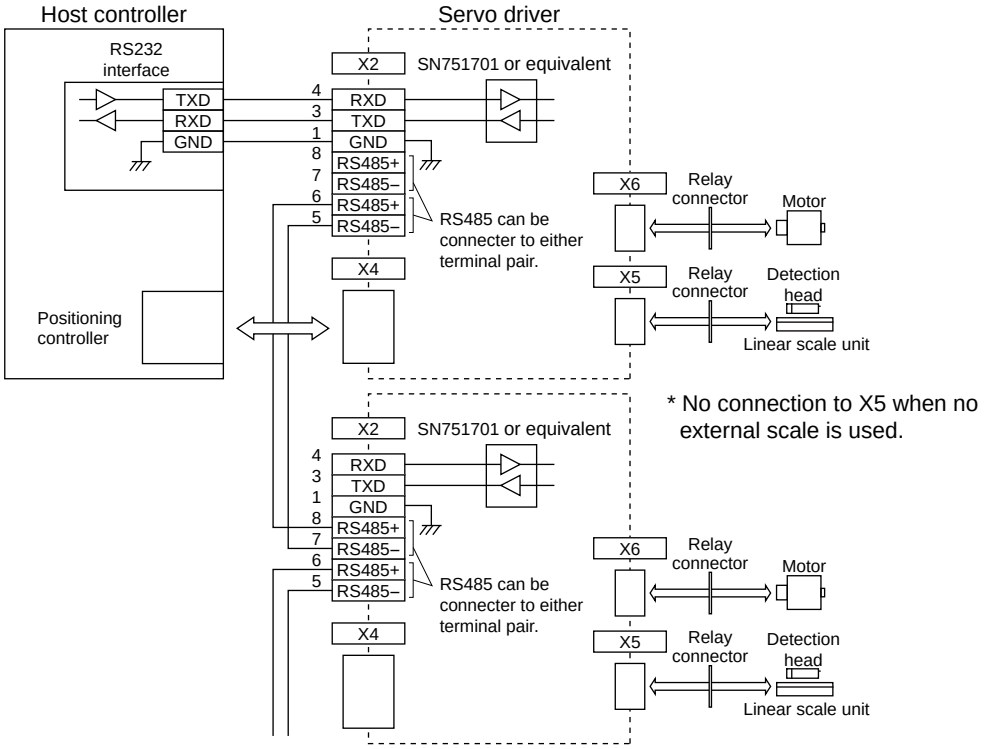
Absolute Specifications

There are 3 connecting methods of the host controller and MINAS-A5 driver as described below, and select a method depending on the interface of the host controller specs or number of axis to be connected. Designate a module ID to Pr5.31 of each MINAS-A5 driver when you connect multiple MINAS-A5 in communication to one host controller as shown below.

[Parameter Pr5.31]

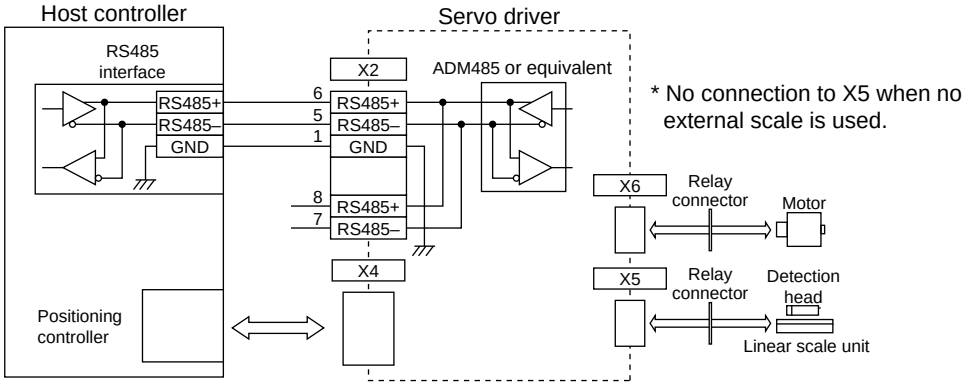
- When you connect each MINAS-A5 to the host separately with RS232 and switch the communication individually, designate 0 to 31 to each MINAS-A5. (Max. 31 axis are connectable.)
- When you connect one MINAS-A5 to the host with RS232 and connect each MINAS-A5 with RS485, designate 0 to the MINAS-A5 connected with the host, and designate 1 to 31 to other MINAS-A5. (Max 32 axis are connectable.)
- When you connect MINAS-A5 to the host with RS485, the host is given module ID of 0, and designate 1 to 31 to MINAS-A5. (Max 31 axis are connectable.)

Configuration of absolute system of scale using RS232 interface



Related page P.7-28 “Connection of Communication Line”

Configuration of absolute system of scale using RS485 interface



Related page P.7-29 “Connection of Communication Line”

Note * Battery for absolute encoder is required to store the multi-turn data into the encoder. Connect the battery between BAT+ and BAT– of the motor.

2. Absolute system

Battery (for Backup) Installation

First Installation of the Battery

After installing and connecting the back-up battery to the motor, execute an absolute encoder setup. Refer to P.7-16, "Setup (initialization) of Absolute Encoder ".
It is recommended to perform ON/OFF action once a day after installing the battery for refreshing the battery.
A battery error might occur due to voltage delay of the battery if you fail to carry out the battery refreshment.

Caution Use the following battery for absolute encoder.
Battery .....Part No. : DV0P2990 (3.6V 2000mAh)
Battery box .....Part No. : DV0P4430

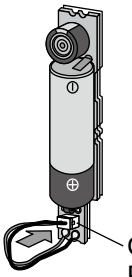
Replacement of the Battery

It is necessary to replace the battery for absolute encoder when battery alarm occurs.
Replace while turning on the control power. Data stored in the encoder might be lost when you replace the battery while the control power of the driver is off.
After replacing the battery, clear the battery alarm. Refer to P.7-25, "How to Clear the Battery Alarm".

Caution When you execute the absolute encoder with the front panel (refer to P.2-79 of Preparation), or via communication (refer to P.7-54), all of error and multi-turn data will be cleared together with alarm, and you are required to execute “Setup (Initialization) of absolute encoder” (refer to P.7-16).

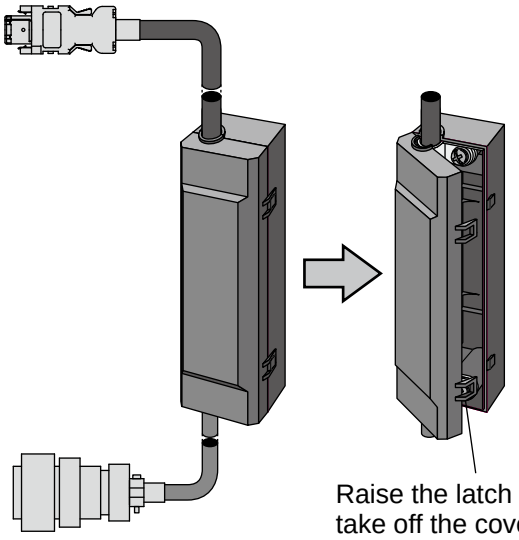
How to Replace the Battery

1) Refresh the new battery.
Connector with lead wire of the battery to CN601 and leave of 5 min. Pull out the connector from CN601 5 min after.



CN601
Pull out after 5 min.
connection

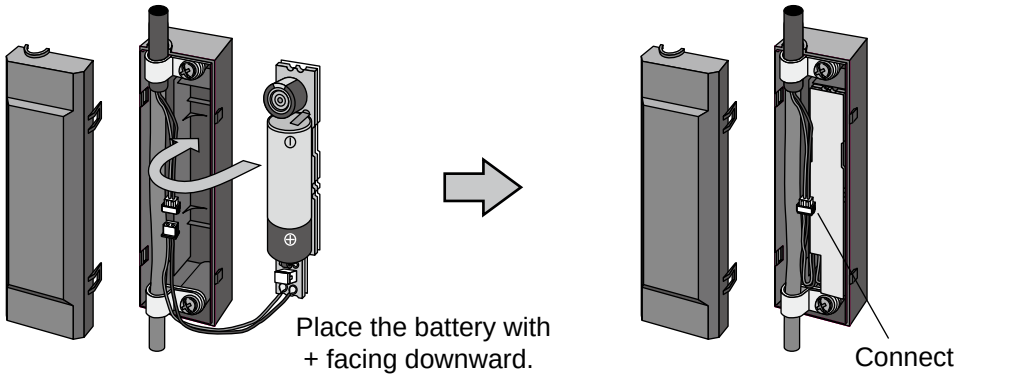
2) Take off the cover of the battery box.



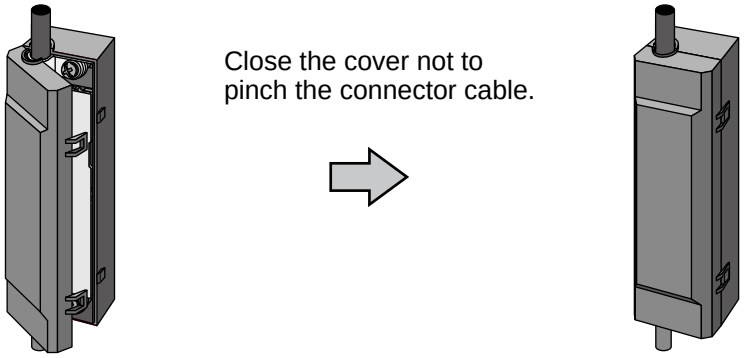
Raise the latch and take off the cover.

Related page • P.7-89 “Battery For Absolute Encoder”

3) Install the battery to the battery box.



4) Close the cover of the battery box.



Caution

- Be absolutely sure to follow the precautions below since improper use of the battery can cause electrolyte to leak from the battery, giving rise to trouble where the product may become corroded, and/or the battery itself may rupture.
  1) Insert the battery with its "+" and "–" electrodes oriented correctly.
  2) Leaving a battery which has been used for a long period of time or a battery which is no longer usable sitting inside the product can cause electrolyte leakage and other trouble. For this reason, ensure that such a battery is replaced at an early date. (As a general guideline, it is recommended that the battery be replaced every two years.)
     - The electrolyte inside the battery is highly corrosive, and if it should leak out, it will not only corrode the surrounding parts but also give rise to the danger of short-circuiting since it is electrically conductive. For this reason, ensure that the battery is replaced periodically.
  3) Do not disassemble the battery or throw it into a fire.
     - Do not disassemble the battery since fragments of the interior parts may fly into your eyes, which is extremely dangerous. It is also dangerous to throw a battery into a fire or apply heat to it as doing to may cause it to rupture.
  4) Do not cause the battery to be short-circuited. Under no circumstances must the battery tube be peeled off.
     - It is dangerous for metal items to make contact with the "+" and "–" electrodes of the battery since such objects may cause a high current to flow all at once, which will not only reduce the battery performance but also generate considerable heat, possibly leading to the rupture of the battery.
  5) This battery is not rechargeable. Under no circumstances must any attempt be made to recharge it.

Caution

The disposal of used batteries after they have been replaced may be subject to restrictions imposed by local governing authorities. In such cases, ensure that their disposal is in accordance with these restrictions.

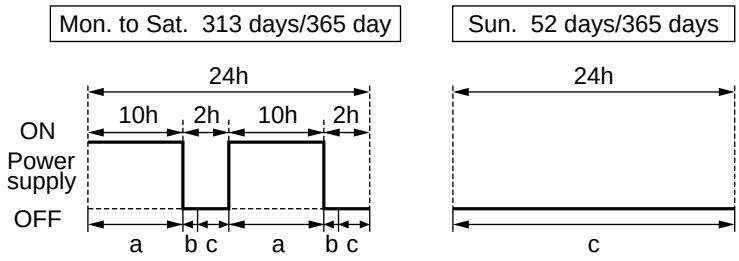
Life of the battery

Following example shows the life calculation of the back-up battery used in assumed robot operation.
2000[mAh] of battery capacity is used for calculation.

Note that the following value is not a guaranteed value, but only represents a calculated value.

Caution The values below were calculated with only the current consumption factored in. The calculations do not factor in electrolyte leakage and other forms of battery deterioration.
Life time may be shortened depending on ambient condition.

1) 2 cycles/day



a : Current consumption in normal mode 3.6[μA]
b : Current consumption at power failure timer mode 280[μA]
* Power failure timer mode...Action mode in time period when the motor can respond to max. speed even the power is off (5sec).
c : Current consumption at power failure mode 110[μA]

Annual consumption capacity =
(10h × a + 0.0014h × b + 2h × c) × 2 × 313 days + 24h × c × 52 days =
297.8[mAh])
Battery life = 2000[mAh]/297.8[mAh] = 6.7 (6.7159) [year]

2) 1 cycle/day

(2nd cycle of the above 1) is for rest.

Annual consumption capacity =
(10h × a + 0.0014h × b + 14h × c) × 313 days + 24h × c × 52 days =
640.6[mAh])
Battery life = 2000[mAh]/630.6[mAh] = 3.1 (3.1715) [year]

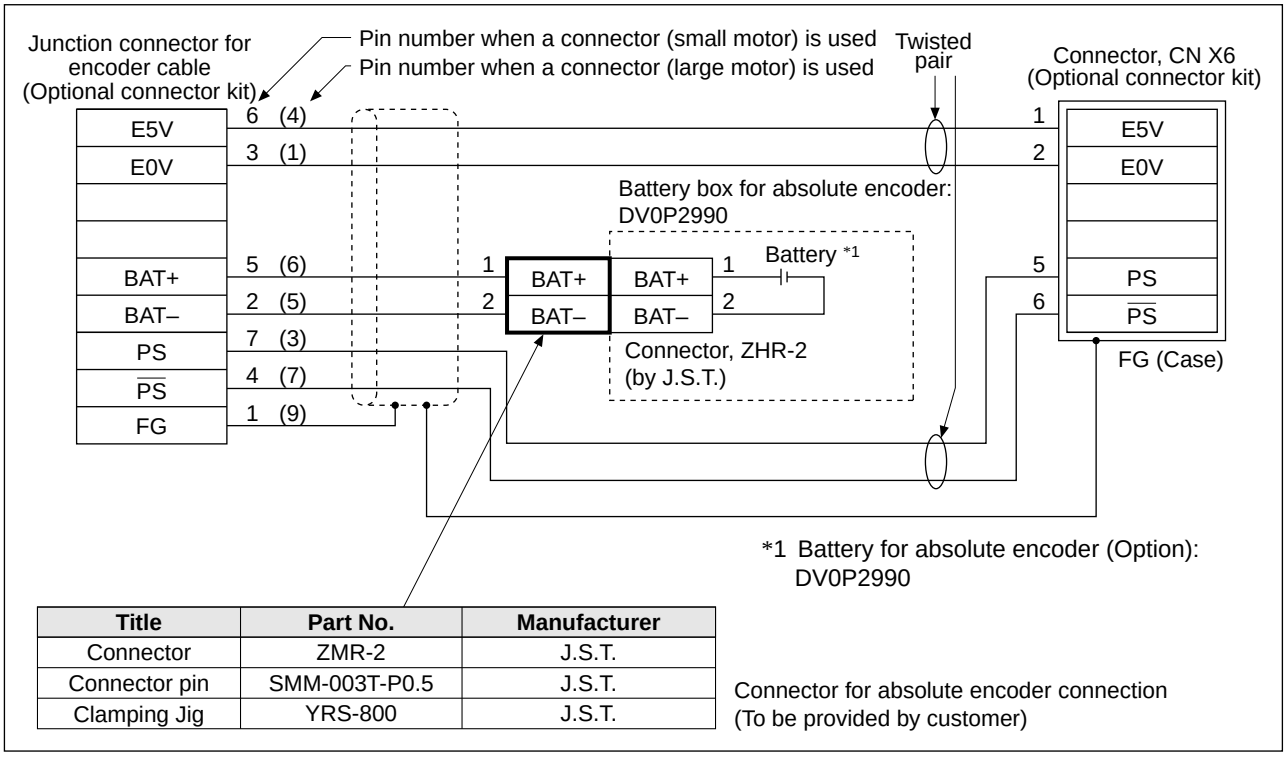
When you make your own cable for 17-bit absolute encoder

When you make your own cable for 17-bit absolute encoder, connect the optional battery for absolute encoder, DV0P2060 or DV0P2990 as per the wiring diagram below. Connector of the battery for absolute encoder shall be provided by customer as well.

Caution Install and fix the battery securely. If the installation and fixing of the battery is not appropriate, it may cause the wire breakdown or damage of the battery.
Refer to the instruction manual of the battery for handling the battery.

- **Installation Place**
  1) Indoors, where the products are not subjected to rain or direct sun beam.
  2) Where the products are not subjected to corrosive atmospheres such as hydrogen sulfide, sulfurous acid, chlorine, ammonia, chloric gas, sulfuric gas, acid, alkaline and salt and so on, and are free from splash of inflammable gas, grinding oil, oil mist, iron powder or chips and etc.
  3) Well-ventilated and humid and dust-free place.
  4) Vibration-free place

- **Wiring Diagram**



Title	Part No.	Manufacturer
Connector	ZMR-2	J.S.T.
Connector pin	SMM-003T-P0.5	J.S.T.
Clamping Jig	YRS-800	J.S.T.

Related page • P.7-89 “Battery For Absolute Encoder”

2. Absolute system

Setup (Initialization) of Absolute Encoder

Absolute multi-turn data will be maintained by the absolute encoder battery.
When operating the machine for the first time after installing the battery to the absolute encoder, clear the encoder data (multi-turn data) to 0 at the origin by following the procedure described below.
Clear the absolute encoder from the front panel (see P.2-79) or PANATERM. Turn off power and then on again.

2. Absolute system

Transferring absolute data

Transfer the absolute data (absolute data of external scale) from the servo driver to the host controller: turn on power and wait until the servo ready output (S-RDY) is turned on, and then start transfer.

Setup of serial communication interface on host controller

• RS232

Baud rate	2400, 4800, 9600, 19200, 38400, 57600, 115200bps
Data	8 bit
Parity	none
Start bit	1 bit
Stop bit	1 bit

The baud rate is set according to Pr5.29 Baud rate setup of RS232 communication.

• RS485

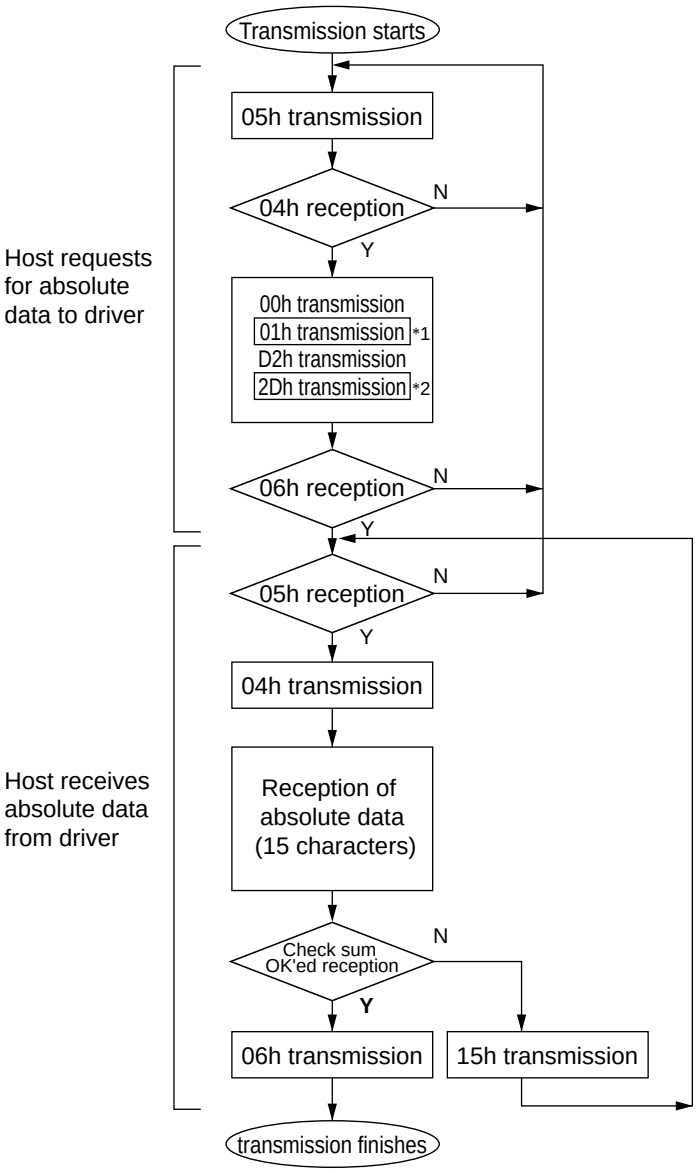
Baud rate	2400, 4800, 9600, 19200, 38400, 57600, 115200bps
Data	8 bit
Parity	none
Start bit	1 bit
Stop bit	1 bit

The baud rate is set according to Pr5.30 Baud rate setup of RS485 communication.

Related page • P.3-30 “Inputs and outputs on connector X4” • P.4-49 “Details of parameter”

RS232 Communication Protocol

Refer to the instruction manual of the host for the transmission/reception method of command.



Data of *1 and *2 are determined by the setup of Pr5.31 “Axis address” of the front panel.

Axis address (example)	Data of *1	Data of *2
0	00h	2Eh
1	01h	2Dh
2	02h	2Ch
3	03h	2Bh
4	04h	2Ah
5	05h	29h
6	06h	28h
7	07h	27h
8	08h	26h
9	09h	25h
10	0Ah	24h
11	0Bh	23h
12	0Ch	22h
13	0Dh	21h
14	0Eh	20h
15	0Fh	1Fh
16	10h	1Eh
17	11h	1Dh
18	12h	1Ch
19	13h	1Bh
20	14h	1Ah
21	15h	19h
22	16h	18h
23	17h	17h
24	18h	16h
25	19h	15h
26	1Ah	14h
27	1Bh	13h
28	1Ch	12h
29	1Dh	11h
30	1Eh	10h
31	1Fh	0Fh

Check sum becomes OK'ed when the lower 8-bit of the sum of the received absolute data (15 characters) is 0.

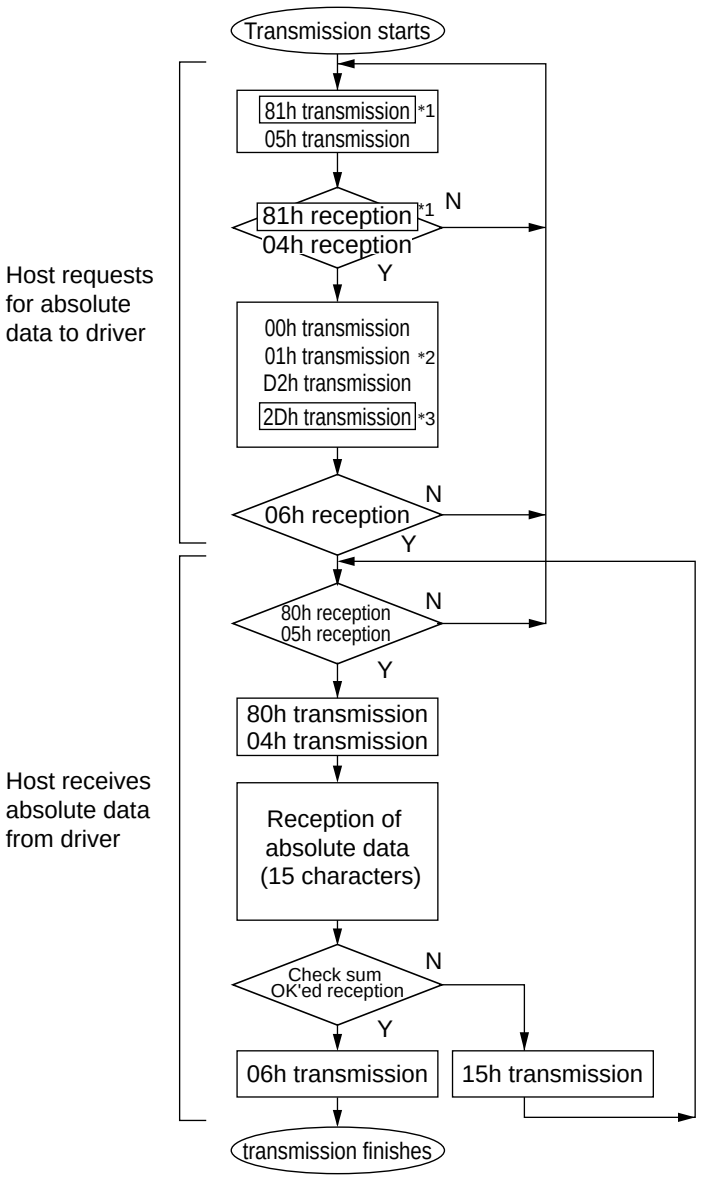
Enter the Pr5.31 value of the driver to which you want to communicate from the host to axis (*1 data) of the command block, and transmit the command according to the RS232 communication protocol. For details of communication, refer to P.7-27, "Communication".

Caution

- **Allow 50ms or longer interval for axis switching when you want to capture multiple axes data.**
- **It is recommended for you to repeat the above communication more than 2 times to confirm the absolute data coincide, in order to avoid mis-operation due to unexpected noise.**

RS485 Communication Protocol

Refer to the instruction manual of the host for the transmission/reception method of command. Following shows the communication example of the driver to Pr5.31 “Axis address = 1”.



Data of *1, *2 and *3are determined by the setup of Pr5.31 “Axis address” of the front panel.

Axis address (example)	Data of *1	Data of *2	Data of *3
0	not usable with RS485 communication		
1	81h	01h	2Dh
2	82h	02h	2Ch
3	83h	03h	2Bh
4	84h	04h	2Ah
5	85h	05h	29h
6	86h	06h	28h
7	87h	07h	27h
8	88h	08h	26h
9	89h	09h	25h
10	8Ah	0Ah	24h
11	8Bh	0Bh	23h
12	8Ch	0Ch	22h
13	8Dh	0Dh	21h
14	8Eh	0Eh	20h
15	8Fh	0Fh	1Fh
16	90h	10h	1Eh
17	91h	11h	1Dh
18	92h	12h	1Ch
19	93h	13h	1Bh
20	94h	14h	1Ah
21	95h	15h	19h
22	96h	16h	18h
23	97h	17h	17h
24	98h	18h	16h
25	99h	19h	15h
26	9Ah	1Ah	14h
27	9Bh	1Bh	13h
28	9Ch	1Ch	12h
29	9Dh	1Dh	11h
30	9Eh	1Eh	10h
31	9Fh	1Fh	0Fh

Check sum becomes OK'ed when the lower 8-bit of the sum of the received absolute data (15 characters) is 0.

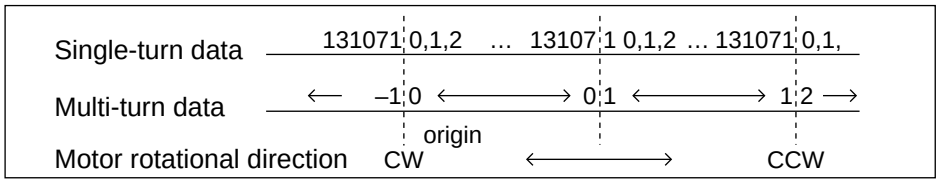
Command from the host will be transmitted to the desired driver based on RS485 transmission protocol. For details of communication, refer to P.7-27, "Communication".

Caution

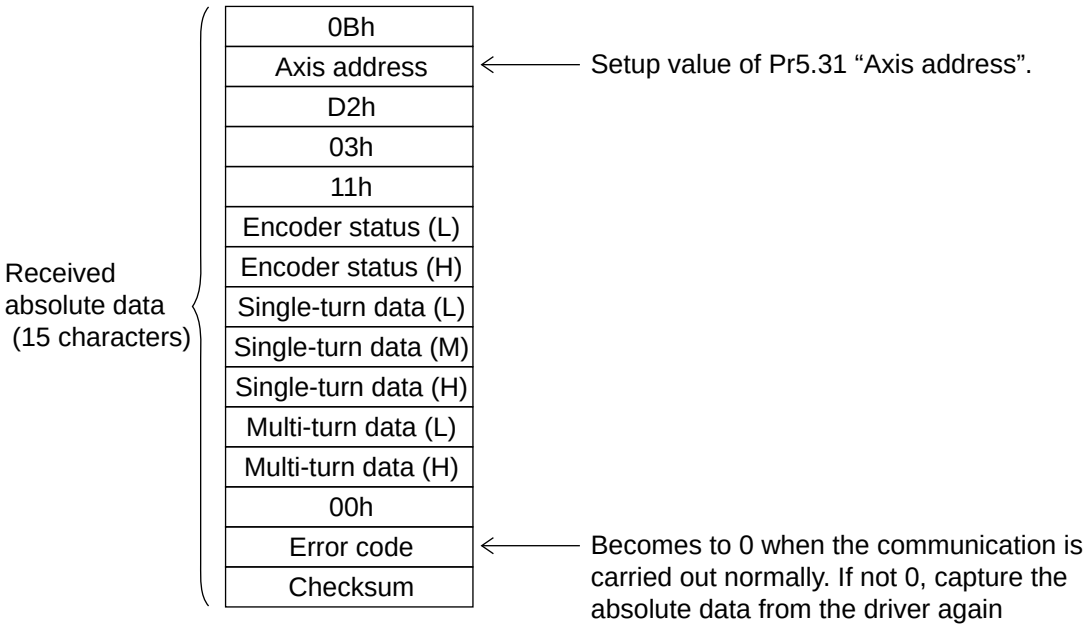
- **Allow 50ms or longer interval for axis switching when you want to capture multiple axes data.**
- **It is recommended for you to repeat the above communication more than 2 times to confirm the absolute data coincide, in order to avoid mis-operation due to unexpected noise.**

Composition of Absolute Data

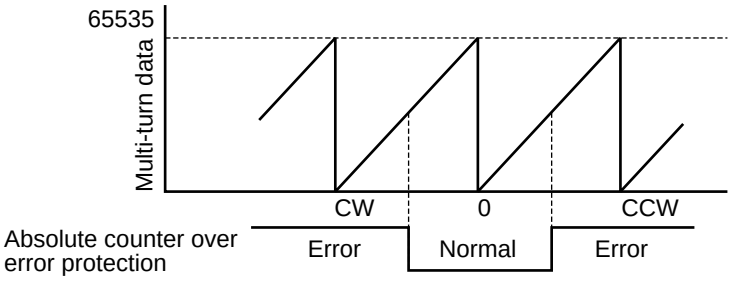
Absolute data consists of singe-turn data which shows the absolute position per one revolution and multi-turn data which counts the number of revolution of the motor after clearing the encoder.



Single-turn data and multi-turn data are composed by using 15-character data (hexadecimal binary code) which are received via RS232 or RS485.



- **Details of multi-turn data**



Single-turn data ← Single-turn data (H)×10000h+Single-turn data (M)×100h+Single-turn data (L)

multi-turn data ← multi-turn data (H) ×100h+multi-turn data (L)

Remarks If the multi-turn data of the above fig. is between 32768 to 65535, convert it to signed date after deducting 65536.

• **Encoder status (L)-----1 represents error occurrence.**

Encoder status (L)							
bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
			0				
(1)	(2)	(3)		(4)	(5)	(6)	(7)

(1) Over-speed → Err42 (Absolute over-speed error protection)
(2) Full absolute status → Err47 (Absolute status error protection)
(3) Counter error → Err44 (Absolute single-turn counter error protection)
(4) Counter overflow → Err41 (Absolute counter over error protection)
(5) Multi-turn error → Err45 (Absolute multi-turn counter error protection)
(6) Battery error → Err40 (Absolute system down error protection)
(7) Battery alarm → Battery alarm

• **Encoder status (L)-----1 represents error occurrence.**

Encoder status (H)							
bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
0	0			0	0	0	0

Battery error

One of the following has occurred.
Battery alarm, multi-turn error, counter overflow, counter error, full absolute status, Counter overflow multi-turn error, battery error or battery alarm

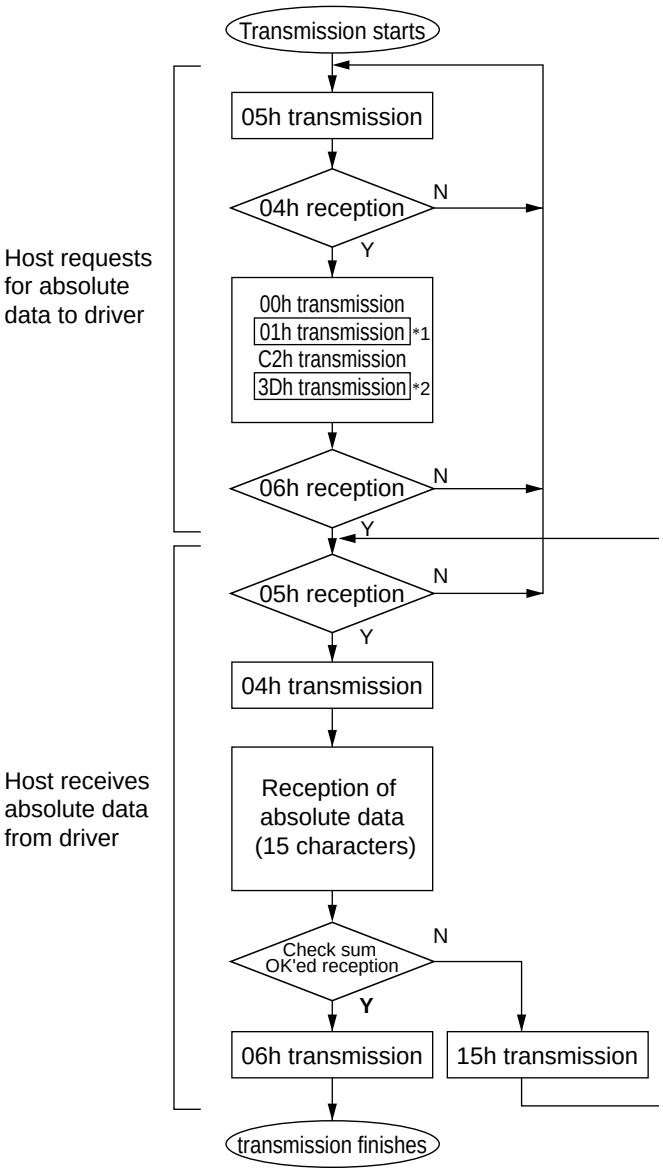
Remarks For detail of the Encoder status, refer to the Encoder specification.

Remarks **• Transmit the absolute data while fixing the motor with brake by turning to Servo-Off.**

Related page • P.6-4 “Protective Function”
• P.7-25 “Display of Battery Alarm”

External scale RS232 communication procedure

Refer to the instruction manual of the host for the transmission/reception method of command. Following shows the communication example of the driver to Pr5.31 “Axis address = 1”.



Data of *1 and *2 are determined by the setup of Pr5.31 “Axis address” of the front panel.

Axis address (example)	Data of *1	Data of *2
0	00h	3Eh
1	01h	3Dh
2	02h	3Ch
3	03h	3Bh
4	04h	3Ah
5	05h	39h
6	06h	38h
7	07h	37h
8	08h	36h
9	09h	35h
10	0Ah	34h
11	0Bh	33h
12	0Ch	32h
13	0Dh	31h
14	0Eh	30h
15	0Fh	2Fh
16	10h	2Eh
17	11h	2Dh
18	12h	2Ch
19	13h	2Bh
20	14h	2Ah
21	15h	29h
22	16h	28h
23	17h	27h
24	18h	26h
25	19h	25h
26	1Ah	24h
27	1Bh	23h
28	1Ch	22h
29	1Dh	21h
30	1Eh	20h
31	1Fh	1Fh

Check sum becomes OK'ed when the lower 8-bit of the sum of the received absolute data (15 characters) is 0.

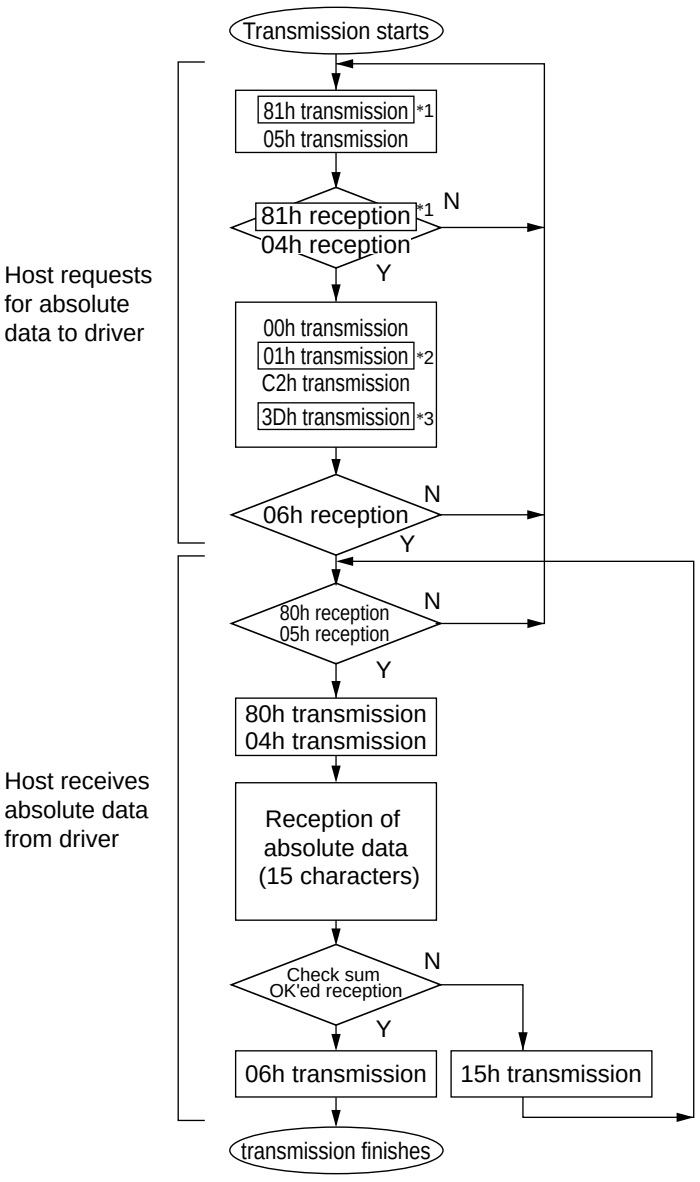
Enter the Pr5.31 value of the driver to which you want to communicate from the host to axis (*1 data) of the command block, and transmit the command according to the RS232 communication protocol. For details of communication, refer to P.7-27, "Communication".

Caution

- **Allow 50ms or longer interval for axis switching when you want to capture multiple axes data.**
- **It is recommended for you to repeat the above communication more than 2 times to confirm the absolute data coincide, in order to avoid mis-operation due to unexpected noise.**

External scale RS485 communication procedure

Refer to the instruction manual of the host for the transmission/reception method of command. Following shows the communication example of the driver to Pr5.31 “Axis address = 1”.



Data of *1, *2 and *3are determined by the setup of Pr5.31 “Axis address” of the front panel.

Axis address (example)	Data of *1	Data of *2	Data of *3
0	Details of parameter		
1	81h	01h	3Dh
2	82h	02h	3Ch
3	83h	03h	3Bh
4	84h	04h	3Ah
5	85h	05h	39h
6	86h	06h	38h
7	87h	07h	37h
8	88h	08h	36h
9	89h	09h	35h
10	8Ah	0Ah	34h
11	8Bh	0Bh	33h
12	8Ch	0Ch	32h
13	8Dh	0Dh	31h
14	8Eh	0Eh	30h
15	8Fh	0Fh	2Fh
16	90h	10h	2Eh
17	91h	11h	2Dh
18	92h	12h	2Ch
19	93h	13h	2Bh
20	94h	14h	2Ah
21	95h	15h	29h
22	96h	16h	28h
23	97h	17h	27h
24	98h	18h	26h
25	99h	19h	25h
26	9Ah	1Ah	24h
27	9Bh	1Bh	23h
28	9Ch	1Ch	22h
29	9Dh	1Dh	21h
30	9Eh	1Eh	20h
31	9Fh	1Fh	1Fh

Check sum becomes OK'ed when the lower 8-bit of the sum of the received absolute data (15 characters) is 0.

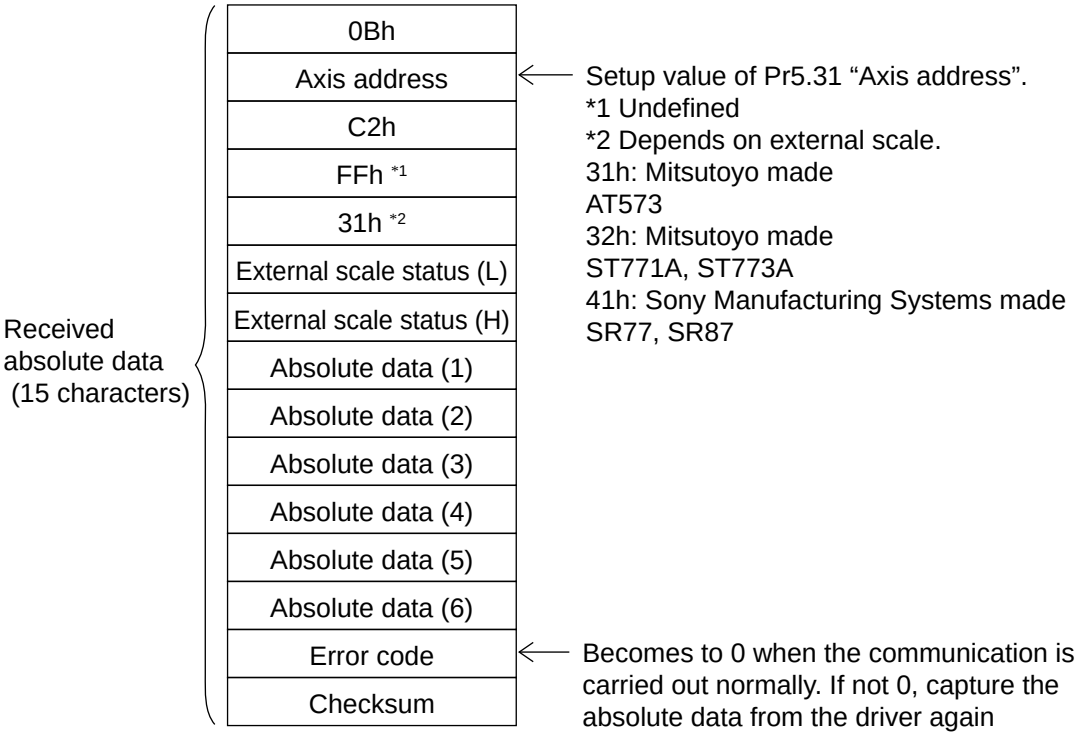
Command from the host will be transmitted to the desired driver based on RS485 transmission protocol. For details of communication, refer to P.7-27, "Communication".

Caution

- **Allow 50ms or longer interval for axis switching when you want to capture multiple axes data.**
- **It is recommended for you to repeat the above communication more than 2 times to confirm the absolute data coincide, in order to avoid mis-operation due to unexpected noise.**

Composition of external scale absolute data

Using 15-character data received through RS232/RS485, organize 1-turn data and multi-turn data.



External scale absolute data

← Absolute data (6)×10000000000h
+Absolute data (5)×100000000h
+Absolute data (4)×1000000h
+Absolute data (3)×10000h
+Absolute data (2)×100h
+Absolute data (1)

Absolute data of external scale is represented as 48 bit number (negative value is represented as two's complement).

Remarks If the multi-turn data of the above fig. is between 32768 to 65535, convert it to signed date after deducting 65536.

• External scale status (L)-----1 represents error occurrence.

External scale status (L)							
bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
			0				
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)

(1) Alarm No. A8 “External scale error alarm”

(2) Alarm No. A8 “External scale error alarm”

(3) Err51.0 “External scale status 5 error protection”

(4) Err51.0 “External scale status 4 error protection”

(5) Err51.0 “External scale status 3 error protection”

(6) Err51.0 “External scale status 2 error protection”

(7) Err51.0 “External scale status 1 error protection”

(8) Err51.0 “External scale status 0 error protection”

• External scale status (L)-----1 represents error occurrence.

External scale status (H)							
bit7	bit6	bit5	bit4	bit3	bit2	bit1	bit0
0	0			0	0	0	0

Logical sum of bit6 and bit 7 of external scale status (L)

Logical sum of bit0 to bit 5 of external scale status (L)

Remarks For detail of the external scale status, refer to the external scale specification.

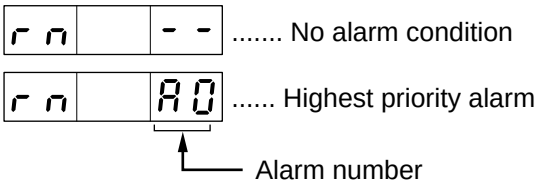
Remarks **• Transmit the External scale absolute data while fixing the motor with brake by turning to Servo-Off.**

Related page
• P.6-4 “Protective Function”
• Next page “Display of Battery Alarm”

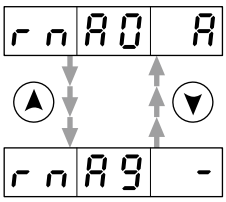
2. Absolute system

Display of Battery Alarm

Following alarm will be displayed when making the front panel to alarm execution mode of monitor mode.



• Press ▲ ▼ to scroll alarm conditions.



• Kinds of alarm

alarm No.	Alarm	Content	Latched time *1
A0	Overload protection	Load factor is 85% or more the protection level.	1 to 10s or ∞
A1	Over-regeneration alarm	Regenerative load factor is 85% or more the protection level.	10s or ∞
A2	Battery alarm	Battery voltage is 3.2 V or lower.	Fixed at ∞
A3	Fan alarm	Fan has stopped for 1 sec.	1 to 10s or ∞
A4	Encoder communication alarm	The number of successive encoder communication errors exceeds the specified value.	1 to 10s or ∞
A5	Encoder overheat alarm	The encoder detects overheat alarm.	1 to 10s or ∞
A6	Oscillation detection alarm	Oscillation or vibration is detected.	1 to 10s or ∞
A7	Lifetime detection alarm	The life expectancy of capacity or fan becomes shorter than the specified time.	Fixed at ∞
A8	External scale error alarm	The feedback scale detects the alarm.	1 to 10s or ∞
A9	External scale communication alarm	The number of successive feedback scale communication errors exceeds the specified value.	1 to 10s or ∞

*1 Alarms can be cleared by using the alarm clear. Because the all existing alarms are kept cleared while the alarm clear input (A-CLR) is kept ON, be sure to turn it OFF during normal operation. Either 1-10s or∞ can be selected by using user parameter.
Exception: Battery alarm is fixed at ∞ because it is latched by the encoder.
Because the end of life alarm means that the life expectancy cannot be extended, the alarm is set at ∞.

How to Clear the Battery Alarm

Replace the battery for absolute encoder when battery alarm occurs according to P.276, "How to Replace the Battery". After replacement, clear the battery alarm in the following 3 methods.

(a) "Connector X4" Connecting Alarm clear input (A-CLR) to COM– for more than 120ms.
(b) Executing the alarm clear function in auxiliary function mode by using the front panel.
(c) Click the "Battery warning" Clear button, after select the "Absolute encoder" tab in the monitor display window by using the PANATERM (option).

Related page
• P.2-75 “Alarm Clear Screen”
• P.3-30 “Inputs and outputs on connector X4”
• P.7-26 “Outline of Setup Support Software, “PANATERM””